Senior Data Scientist Interview Questions

ML Interview Preparation Notes

Prepared from the LinkedIn post by Pooja Kumawat

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Note: The LinkedIn post provides ten questions, not official answers. The answers below are suggested interview frameworks. Explain the reasoning and trade-offs rather than memorizing them verbatim.

Q1. End-to-end customer churn prediction

Question. Design an end-to-end ML solution to predict churn at scale, covering features, model selection, validation, deployment, and monitoring.

Hinglish notes. Churn ko precisely define karo, e.g. next 30 days mein cancellation. Features mein activity recency/frequency, spending trends, complaints, failed payments, tenure aur support interactions ho sakte hain. Logistic regression baseline banao, phir LightGBM/XGBoost/CatBoost compare karo. Temporal validation use karo; future information leak nahi honi chahiye. Batch/online deploy karo aur PR-AUC, recall, calibration, latency, drift aur business impact monitor karo. Risk prediction aur intervention se customer ko bachane ki causal ability alag cheezein hain.

Final answer (English). I would define a clear churn horizon, build point-in-time features, establish a logistic regression baseline, and compare boosted-tree models. I would use temporal validation, select metrics based on retention costs, deploy a monitored pipeline, and measure incremental retained revenue rather than accuracy alone.

Q2. 98% accuracy but poor production results

Question. Diagnose poor production performance and select better metrics for imbalanced classification.

Hinglish notes. Agar 1000 mein 980 negative aur 20 positive hain, har case ko negative predict karne se bhi 98% accuracy milti hai, lekin recall zero hota hai. Confusion matrix, precision, recall, PR-AUC, ROC-AUC, calibration aur threshold evaluate karo. Production mein data drift, label delay, missing features, training-serving skew, aur incorrect joins check karo. Threshold validation data aur business costs ke basis par choose karo.

Final answer (English). I would inspect the confusion matrix and class distribution, evaluate precision, recall, PR-AUC, calibration, and business costs, then tune the decision threshold on validation data. I would also investigate drift, data quality, and training-serving skew before assuming the model itself is at fault.

Q3. Data leakage

Question. Explain leakage during feature engineering, cross-validation, or time-series modeling, and how to prevent it.

Hinglish notes. Leakage tab hota hai jab future ya validation information training ko indirectly mil jaati hai. Examples: churn ke baad generate hua cancellation field, full dataset par scaler fit karna,

ya future records se past predict karna. Split pehle karo, transformations ko Pipeline/CV folds ke andar fit karo, point-in-time feature availability enforce karo, aur temporal/group splits choose karo. Pipeline galat split ko fix nahi karti.

Final answer (English). I would prevent leakage by splitting data before preprocessing, fitting transformations within training folds, enforcing point-in-time feature availability, and using temporal or group-aware validation when appropriate. I would audit suspicious features and unusually strong validation scores.

Q4. Interpretability for XGBoost or deep learning

Question. Explain predictions to business and regulatory stakeholders using SHAP, feature importance, and their limitations.

Hinglish notes. Global feature importance batati hai model overall kis information ka use karta hai; local explanation ek prediction ke contributors batati hai. SHAP local attribution de sakta hai aur global summaries mein aggregate ho sakta hai. Correlated features importance share ya distort kar sakte hain; explanations model behavior explain karti hain, causality nahi. Stability, subgroup behavior, counterfactual plausibility aur documentation check karo. High-stakes decisions mein human review aur simpler surrogate/baseline bhi consider karo.

Final answer (English). I would combine global feature importance with local SHAP explanations, validate explanation stability across samples and groups, and communicate limitations clearly. These methods describe model associations, not causal effects, so I would pair them with domain checks, fairness analysis, and documented human oversight where needed.

Q5. Causal inference and experimentation

Question. A recommendation algorithm allegedly increased revenue by 12%. Determine whether the algorithm caused the increase.

Hinglish notes. Before-after comparison alone enough nahi hai: seasonality, promotions, traffic mix aur selection bias result explain kar sakte hain. Best option randomized A/B test hai: eligible users ko random treatment/control assign karo, primary metric (e.g. revenue per eligible user) aur guardrails pehle define karo, sample size/power calculate karo, experiment duration aur stopping rules predefine karo. Analyze intention-to-treat with confidence intervals. Randomization impossible ho to difference-in-differences, matching, or synthetic controls use ho sakte hain, but assumptions state karni hongi.

Final answer (English). I would run a randomized controlled A/B test with pre-specified metrics, adequate power, guardrails, and an intention-to-treat analysis. I would report the effect size and confidence interval and check for sample-ratio mismatch and implementation issues. If randomization is infeasible, I would use a defensible quasi-experimental design and state its assumptions.

Q6. Production model drift

Question. Performance declines over three months. Distinguish data drift, concept drift, and target-distribution changes; design monitoring and retraining.

Hinglish notes. Data drift: input features ka distribution badla. Prior/target shift: class prevalence P(Y) badli. Concept drift: P(Y|X) relationship badla. Ye changes overlap bhi kar sakte hain. Feature distributions, missingness, prediction scores, calibration aur delayed labels monitor karo; cohorts/segments mein metrics compare karo. Retraining sirf drift alarm par blindly nahi: label-backed degradation, business impact, retraining evaluation, rollback aur champion-challenger use karo.

Final answer (English). I would monitor feature and prediction distributions, delayed-label performance, calibration, and segment-level metrics. I would distinguish input drift from changes in class prevalence and in the relationship between features and outcomes. Retraining would be triggered by validated degradation and followed by offline evaluation, canary deployment, and rollback safeguards.

Q7. Training and inference for billions of records

Question. Design scalable Spark-based training and inference, considering latency, cost, and performance.

Hinglish notes. Distributed ingestion/ETL ko partitioned data par run karo, schema/data-quality checks aur reproducible feature definitions rakho. Spark SQL/DataFrames use karo; skew, shuffles, joins, partition sizing aur caching carefully optimize karo. Training ke liye suitable distributed learner ya sampled/aggregated training strategy choose karo. Batch inference throughput/cost ke liye; online serving low latency ke liye. Feature consistency, model registry, idempotency, retries, access control, and monitoring essential hain. Pehle benchmark and SLA define karo.

Final answer (English). I would build partitioned, fault-tolerant data pipelines with reproducible feature definitions and distributed training suited to the model. I would choose batch or online inference according to latency SLAs, optimize shuffles and data skew, and benchmark cost, throughput, and model quality at realistic scale.

Q8. Time-series demand forecasting

Question. Forecast thousands of products with intermittent sales, seasonality, and changing market conditions.

Hinglish notes. Naive/seasonal-naive baseline se start karo. Dense, seasonal products ke liye ETS, ARIMA, gradient boosting with lag/calendar/promo features, or global models compare karo. Intermittent demand ke liye Croston-family methods ya suitable probabilistic models consider karo. Rolling-origin backtesting use karo; random split nahi. WAPE/MAE/RMSE/quantile loss business context ke mutabik choose karo; MAPE zero demand par problematic hai. Prediction intervals aur calibration se uncertainty report karo.

Final answer (English). I would segment products by demand pattern, establish naive baselines, and compare classical, intermittent-demand, and global ML approaches. I would use rolling-origin backtests, metrics appropriate for zero-heavy demand, and probabilistic forecasts or prediction intervals to quantify uncertainty.

Q9. AUC improves but profitability does not

Question. Align objective, threshold, and evaluation with business profitability.

Hinglish notes. AUC ranking measure hai; woh directly profit optimize nahi karta. Har prediction ke saath false positive/false negative ki cost, intervention cost, margin aur capacity consider karo. Expected utility/profit ke basis par threshold choose karo, calibration assess karo, aur decision policy ko validation data par optimize karo. Final impact randomized experiment se validate karo. AUC improvement useful ho sakta hai, but business metric guarantee nahi.

Final answer (English). I would translate model outcomes into expected business value, incorporate false-positive and false-negative costs, intervention costs, and operational constraints, then optimize the decision threshold or policy on validation data. I would assess calibration and confirm incremental profitability through an experiment.

Q10. Interpretable model versus complex ensemble

Question. Resolve a team disagreement between a simple interpretable model and a complex ensemble.

Hinglish notes. Debate ko opinion se evidence par lao. Define requirements: performance, calibration, latency, cost, robustness, fairness, interpretability, auditability, and maintenance. Same split aur same metrics par baseline logistic/simple tree vs ensemble compare karo. Agar ensemble ka gain small hai aur risk/compliance high, simple model better ho sakta hai. Agar gain material hai, explanations, monitoring, fallback aur governance ke saath ensemble choose kar sakte ho. Decision record karo and stakeholders se align karo.

Final answer (English). I would define decision criteria with stakeholders and compare both approaches on identical data and business-relevant metrics, including calibration, latency, cost, robustness, and interpretability. I would recommend the simplest model that meets requirements unless the ensemble provides a material, validated benefit that justifies its additional complexity and governance burden.